

Glaucoma: mechanism, surgery and postoperative reasoning

Parts 1 and 2 | Examine the pathway before naming the glaucoma

Angle anatomy identifies which treatment mechanisms are available.

Appositional contact

May open with appropriate indentation

Permanent synechial adhesion

Remains closed at the adherent site

Use gentle technique and interpret the whole angle.

Original conceptual map: static contact and permanent adhesion require different interpretation.

Gonioscopy complements pressure and nerve assessment. Lighting and inadvertent pressure can alter the observed angle; indentation can help distinguish appositional closure from synechial adhesion when appropriate. Record pigmentation, abnormal vessels, recession and membrane-related change. A normal central chamber depth does not replace examination of the peripheral angle.

Progression requires reliable longitudinal evidence

A damaged field and a progressing field are not the same statement.

Check	Reason
Reliability and learning effects	Avoid progression from test noise
Repeatable location of change	Supports a real functional trend
Central versus peripheral sampling	Test grid must match the remaining field
Structure-function agreement	Discordance should prompt reassessment

Interpret the series rather than one printout. Media opacity, refractive error and other neurological or retinal disease can alter fields. Advanced structural measurements may reach a floor while functional change continues. The rate and location of loss determine urgency alongside life expectancy, visual needs and treatment burden.

The target is a clinical hypothesis

Reassess it when observed progression contradicts its adequacy.

Factor	Target implication
Existing damage and central threat	Less tolerance for further loss
Rate of progression	May justify stronger lowering
IOP pattern and adherence	Clinic readings may miss inadequate exposure
Adverse effects and preferences	Changes acceptable treatment burden

Primary open-angle glaucoma requires exclusion of relevant secondary causes, not merely an open angle. Consider longitudinal optic nerve and functional evidence. A pressure within a population reference range can still be too high for a particular nerve.

Treatment success is preservation of useful function, not achievement of an isolated number.

Laser acts on an accessible trabecular target

It cannot reopen every synechially closed or membrane-covered pathway.

Visible trabecular target



Original section: laser requires access to the trabecular meshwork it is intended to treat.

Assess angle visibility, pigmentation, inflammation and the intended goal before trabeculoplasty. Response and durability vary; arrange pressure surveillance and a plan for insufficient benefit. Repeat treatment can be relevant in selected eyes, but the label open angle does not guarantee equivalent effectiveness across all secondary mechanisms.

Evidence informs an operation, not an automatic rule

Separate functional outcomes from the chosen primary endpoint.

TAGS five-year finding	Interpretation
Lower IOP with primary trabeculectomy	Supports greater pressure lowering
Better visual-field preservation	Clinically relevant disease-control outcome
No detected vision-specific QoL difference	Different endpoint from IOP
Newly diagnosed advanced OAG population	Do not generalise to every secondary glaucoma

TAGS compared primary trabeculectomy with medical management in newly diagnosed advanced disease. Discuss the evidence alongside surgical risk, monitoring capacity and patient preferences. Similar observed safety in a trial does not mean surgery has no complications, and a nonsignificant quality-of-life difference does not negate a field-preservation benefit.